PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : GASEOUS STATE (Question Bank # 09)

MM : 69 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. If 250 mL of N_2 over water at 30°C and a total pressure of 740 torr is mixed with 300 mL of Ne over water at 30°C and a total pressure of 780 torr, what will be the total pressure if the mixture is in a 500 mL vessel over water at 30°C

(Given : Vapour pressure (Aqueous tension) of H_2O at 30°C is 40 torr. Assume volume of H_2O (*l*) is negligible in final vessel)

(A) 760 torr (B) 820 torr (C) 780 torr (D) 840 torr

2. If the number of molecules of SO_2 (M = 64) effusing through an orifice of unit area of cross section in unit time at 0°C and 1 atm pressure is **'n'**. The number of He molecules (M = 24) effusing in unit time under similar conditions at 273°C and 0.25 atm is :

(A) $\frac{n}{\sqrt{2}}$ (B) $\frac{n}{2}$ (C) 2n (D) $\sqrt{2}$ n

3. The most probable kinetic energy of molecules in a sample of ideal gas is

(A) KT (B) $\frac{1}{2}$KT (C) $\frac{3}{2}$KT (D) $\frac{1}{3}$KT

4. An open vessel at 27° C is heated untill 3/8th of air in it has been expelled. Assuming that the volume remain constant, find the temperature at which the vessel was heated :

(A) 307° C (B) 107° C (C) 480° C (D) 207° C

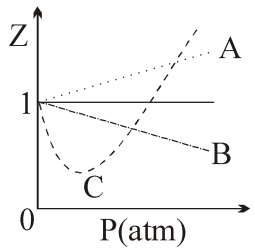
5. The density of a gaseous substance at 1 atm pressure and 773 K is 0.4 g/L. If the molecular weight of the substance is 30, then dominant forces existing among gas molecules is :

(A) repulsive (B) attractive (C) both (D) none

6. 0.5 mol of a diatomic gas present in 10 L vessel at certain temperature exert a pressure of 0.96 atm. Under similar conditions an ideal gas exerted 1.0 atm pressure. If volume of gas molecules is negligible then, the value of vander waals constant "a" (in atm L^2/mol^2) is :

(A) 4 (B) 16 (C) 32 (D) 8

7. The given graph represents the variation of Z (compressibility factor = $\frac{PV}{nRT}$) versus P, for three real gases A, B and C. Identify the only **INCORRECT** statement.



(A) for the gas A, a = 0 and its dependence on P is linear at all pressure
(B) for the gas B, b = 0 and its dependence on P is linear at all pressure
(C) for the gas C, which is typical real gas for which neither a nor b = 0. By knowing the minima and the point of intersection, with Z = 1, a and b can be calculated.
(D) At high pressure, the slope is positive for all real gases A, B and C.

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8. One litre of a gas at 300 atm and 473 K is compressed to a pressure of 600 atm and 273 K. The compressibility factors are 1.072 and 1.375 respectively at initial and final states. What will be the final volume :

(A) 307.8 mL (B) 370.1 mL

(C) 22.4 mL (D) 740.2 mL

(ii) One or more options correct Type

This section contains **4 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

9. Select the correct statement for a real gas

(i) The behaviour of the gas become more ideal when pressure is very low.

(ii) Pressure of gas for ideal behaviour depends upon nature of the gas and its temperature.

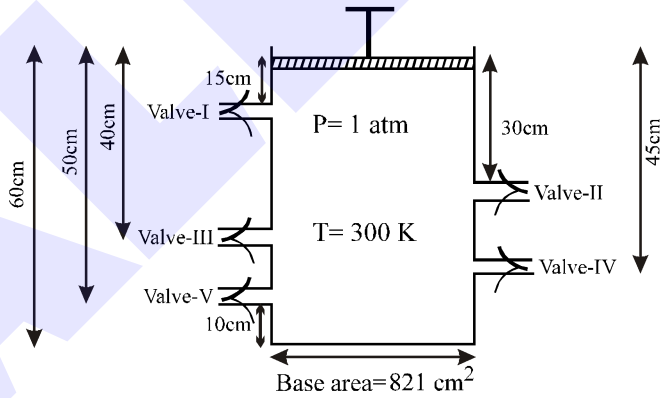
(iii) The temperature at which a real gas obeys ideal gas law over an appreciable range of pressure in low pressure range is called Boyle's temperature.

(A) Only (i), (ii) (B) Only (i)

(C) Only (ii), (iii) (D) (i), (ii), (iii)

10. A container fitted with frictionless massless piston consist of five **valves–I, II, III, IV** and **V.** These valves open automatically if pressure exceed over **1.5, 2.2, 2.5, 4.4** and **4.8 atm** respectively. Under the given initial conditions (mentioned in given diagram) system is in state of equilibrium. Piston is now pressed in downward direction very slowly.

[Note: Consider the diameter of valve tube negligible and temperature remain constant.]



Select the correct option(s):

(A) **Valve –II** will be opened first

(B) As the piston crosses the valve which will be opened first, the remaining number of moles in container are $\frac{5}{3}$.

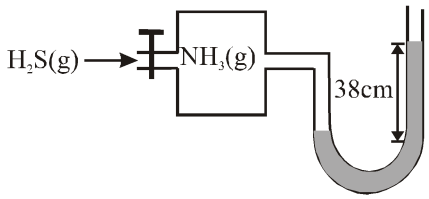
(C) **Valve–V** will be the second valve which open

(D) Number of moles will zero as piston crosses **Valve–V**

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11. A closed container of volume 4.92 litre contains NH_3 gas at 27°C as shown:



H_2S gas is passed inside the container at a rate of 0.01 mole/sec, such that no gas is coming out from the container. The reaction take place in the container is

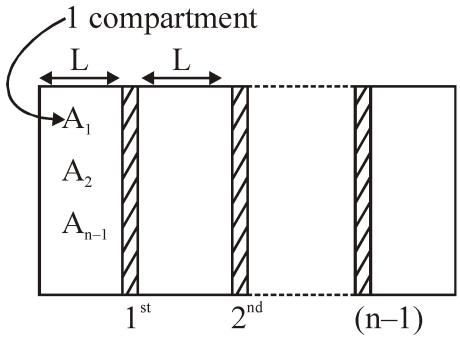
$$2NH_3(g) + H_2S(g) \xrightarrow{100\%} (NH_4)_2S(s)$$

(Assume that there is no time lag in between H_2S gas passed and $(NH_4)_2S$ formed.)

The time at which the difference in Hg column is equal to 38 cm.

(A) 0 sec. (B) 20 sec. (C) 10 sec (D) 25 sec.

12. A very long rectangular box is divided into **'n' equal compartments** with (n – 1) **fixed SPM** (semi permeable membrance) numbered from 1 to (n – 1) [n is unknown] as shown. The gases are initially present in only 1 compartment & all gases have **same moles** & after substantial time ratio of $\mathbf{P_{A_4}}$ in 3rd **compartment** to $\mathbf{P_{A_{n-1}}}$ in 1st **compartment** is 3 (where P_{A_4} & $P_{A_{n-1}}$ represents partial pressure of A_4 & A_{n-1} respectively).



Select the correct option(s) based on given condition.

Condition-1 : Gases can pass through only those SPM whose number is less than to their subscript (like A_2 can pass through 1st SPM only, A_3 through 1st & 2nd & so on).

Condition-2 : Gases can pass through only those SPM whose number is less than or equal to their subscript (like A_1 can pass through 1st SPM only, A_2 through 1st & 2nd & so on).

(A) The value of n is 15 if **condition-2** is followed

(B) The value of n is 13 if **condition-1** is followed

(C) After substantial time ratio of n_{A_3} in 2nd compartment to n_{A_5} in 4th compartment is 1.5 if **condition-2** is followed (where n_{A_3} and n_{A_5} are number moles of A_3 and A_5 moles respectively)

(D) After substantial time ratio of P_{A_6} and in 6th compartment to P_{A_1} in 1st compartment is 2/7 if **condition-1** is followed

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(iii) Paragraph Type

This section contains **1 paragraph** , describing theory, experiment, data etc. **3 questions** relate to one paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Pagragraph for question nos. 13 to 15

Recent communications with the inhabitants of Neptune have revealed that they have a Celsius type temperature scale, but based on the melting point (0°N) and boiling point (100°N) of their most common substance, hydrogen. Further communications have revealed that the Neptunians know about perfect gas behaviour and that they find in the limit of zero pressure, the value of $\mathbf{PV_m}$ is 28L - atm at 0°N and 40L - atm at 100°N.

13. What is the value of the absolute zero of temperature on their temperature scale?

(A) 233.33°C (B) – 233.33°C

(C) – 233.33°N (D) 233.33°N

14. What is the value of universal gas constant?

(A) 0.082L - atm/ K - mol (B) 0.12L- atm/ K- mol

(C) 0.4L - atm/K-mol (D) infinite

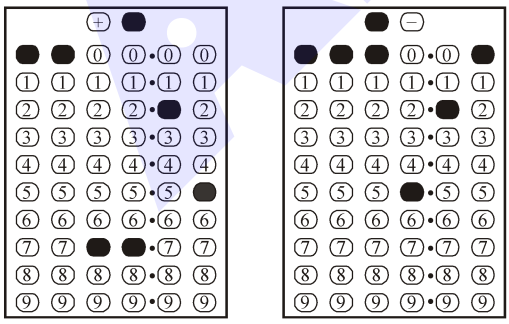
15. What is the volume occupied by 2 mole of a perfact gas at 2 atm and 66.67°N?

(A) 36 L (B) 24.6 L (C) 20.0 L (D) 8.0 L

SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

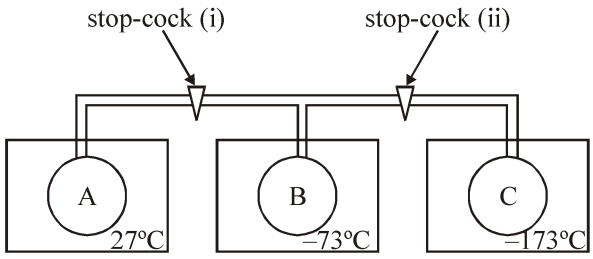
Zero Marks : 0 In all other cases.

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1. 500 ml of a vessel contains an ideal gas saturated with water vapours at 3.2 atm and 47ºC. Now the entire mass is transferred into vessel of 1800 ml capacity maintained at 87ºC. The final pressure **(in atm)** becomes (vapour pressure of water at 47º and 87º C are 0.2 atm and 0.6 atm, respectively)

2.



The above apparatus consists of three temperature jacketed 12.315 litre bulbs connected by stop cocks. Bulbs 'A' consist of $NH_{3(g)}$, $CO_{2(g)}$ & $N_{2(g)}$ at 27ºC and a total pressure of 60 atm. Bulb 'B' and 'C' are empty and maintained at temperature of –73ºC and –173ºC respectively. Initially stop-cocks are closed and volume of the connecting lines is zero.

Given :

(i) $NH_{3(g)}$ liquifies at –20ºC, $CO_{2(g)}$ liquifies at –80ºC and $N_{2(g)}$ liquifies at –180ºC.

(ii) Vapour pressure of $NH_{3(\ell)}$ at –73ºC = 1 atm

(iii) Vapour pressure of $NH_{3(\ell)}$ at –173ºC is negligible

(iv) Vapour pressure of $CO_{2(\ell)}$ is negligible

(v) On opening stop-cock (i) only, the pressure in containers becomes 17 atm and, on opening both stop-cocks pressure decreases to 4 atm

what is the value of $(n_{NH_3} + n_{CO_2} - n_{N_2})$ in the initial mixture ?

3. A barometer reads 75 cm Hg, when the atmospheric pressure is 80 cm Hg. The same barometer reads 84 cm Hg, when the atmospheric pressure is 90 cm Hg. What should be the reading of barometer when the atmospheric pressure is 70 cm Hg. (Given : $\sqrt{199^2 - 4 \times 8760} = 68$)

If the answer is x cm Hg, fill OMR the value of $\left(\frac{8x}{131}\right)$.

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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **four statements** (A, B, C and D) given in **Column I** and 4 statements (P, Q, R, and S) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question, against statement B, darken the bubbles corresponding to Q and R in the ORS. **8(0)**

1. Match gases under specific conditions listed in Column I with their properties / laws in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

	Column I		**Column II**
(A)	Hydrogen gas (P = 200 atm, T = 273 K)	(P)	Compressibility factor ≠ 1
(B)	Hydrogen gas (P ~ 0, T = 273 K)	(Q)	Attractive forces are dominant
(C)	CO_2 (P = 1 atm, T = 273 K)	(R)	PV = nRT
(D)	Real gas with very large molar volume	(S)	P (V– nb) = nRT

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ANSWER KEY

SECTION–I : (i) Only One option correct Type: 3(–1)

1.	Ans. (D)	2.	Ans. (A)	3.	Ans. (B)	4.	Ans. (D)
5.	Ans. (A)	6.	Ans. (B)	7.	Ans. (D)	8.	Ans. (B)

(ii) One or more options correct Type : 4(–1)

9.	Ans. (D)	10.	Ans. (B,C)	11.	Ans. (A,C,D)	12.	Ans. (A,B,C)

(iii) Paragraph Type: 3(0)

13.	Ans. (C)	14.	Ans. (B)	15.	Ans. (A)

SECTION-II: (Integer Second Decimal) : 4(0)

1.	Ans. (1.00)	2.	Ans. (8.00)	3.	Ans. (4.00)

SECTION–IV : Matrix-Match Type : 8(0)

1. Ans. (A) P, S; (B) R; (C) P, Q; (D) R

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